

A machine learning model based on high-frequency ultrasound for differentiating benign and malignant skin tumors

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Abstract

Aim: This study aims to explore the potential of machine learning as a non-invasive automated tool for skin tumor differentiation. **Material and methods:** Data were included from 156 lesions, collected retrospectively from September 2021 to February 2024. Univariate and multivariate analyses of traditional clinical features were performed to establish a logistic regression model. Ultrasound-based radiomics features are extracted from grayscale images after delineating regions of interest (ROIs). Independent samples t-tests, Mann-Whitney U tests, and Least Absolute Shrinkage and Selection Operator (LASSO) regression were employed to select ultrasound-based radiomics features. Subsequently, five machine learning methods were used to construct radiomics models based on the selected features. Model performance was evaluated using receiver operating characteristic (ROC) curves and the Delong test. **Results:** Age, poorly defined margins, and irregular shape were identified as independent risk factors for malignant skin tumors. The multilayer perception (MLP) model achieved the best performance, with area under the curve (AUC) values of 0.963 and 0.912, respectively. The results of DeLong's test revealed a statistically significant discrepancy in efficacy between the MLP and clinical models ($Z=2.611$, $p=0.009$). **Conclusion:** Machine learning-based skin tumor models may serve as a potential non-invasive method to improve diagnostic efficiency.

Keywords: Machine Learning; High-Frequency Ultrasound; Skin Tumor

Introduction

Skin tumors encompass a variety of abnormal proliferative lesions originating from the epidermis, dermis, subcutaneous tissue, and skin appendages, which are clinically classified as benign or malignant [1]. Benign skin tumors include compound nevus, seborrheic keratosis, pilomatricoma, hemangioma, and epidermoid cysts, while malignant skin tumors comprise basal cell carcinoma, squamous cell carcinoma, and melanoma. Many endogenous and exogenous factors contribute to making

malignant skin tumors among the most aggressive types of cancer [2-4]. Skin cancer, characterized by invasive growth, may metastasize and lead to mortality [5]. However, some early-stage malignant skin tumors are difficult to distinguish from benign lesions or dermatological conditions, so a non-invasive examination is prudent before choosing an invasive biopsy [6].

Noninvasive clinical examinations primarily include dermoscopy and high-frequency ultrasound (HFUS), both of which are widely utilized [7]. Optical Coherence Tomography (OCT) and Reflectance Confocal Microscopy (RCM) are both effective non-invasive techniques for the high-sensitivity detection of skin tumors. Similar to OCT and RCM, dermoscopy significantly improves the diagnostic accuracy of skin cancers (particularly melanoma). However, each of these methods has its own limitations. Beyond the considerations of equipment accessibility and the level of training among clinicians, which can impact diagnostic outcomes and the infor-

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mation available about lesions – the limited penetration depth of these techniques remains a critical drawback. This limitation hinders their ability to evaluate lesion internal structures, tumor depth, invasiveness, and vascular distribution [8-10]. HFUS has been shown to aid in the accurate differentiation of benign and malignant skin diseases, with significant advantages over other imaging modalities [11]. However, widely used non-invasive diagnostic approaches and histopathological evaluation in clinical practice depend on the expertise and experience of ultrasound physicians, introducing subjectivity.

The emergence of artificial intelligence (AI) algorithms has transformed the field of medical imaging, and automated tools for screening and early diagnosis of skin tumors have the potential to enhance the diagnostic efficiency and accuracy of radiologists [12]. Machine learning technologies for HFUS analysis have matured and have been successfully applied to identifying and diagnosing thyroid and breast nodules, as well as assessing lymph node abnormalities [13-15].

In this study, we developed five models capable of performing basic classification tasks (i.e., distinguishing malignant from benign skin tumors) using different machine-learning algorithms and we evaluated their performance. Furthermore, we compared the best machine learning model from this study with a clinical model developed using clinical and ultrasound information to validate its clinical application value.

Material and methods

Patients

This study retrospectively collected information from 143 patients with skin tumors treated at the First Affiliated Hospital of Dalian Medical University between September 2021 and February 2024. Each patient had one or more skin tumors examined. All cases included in the study had complete ultrasound examination records and postoperative histopathological biopsy data. Demographic characteristics of the patients (age, gender) and available tumor characteristics (location) were compiled, with histopathological results serving as a reliable reference for describing the type of skin tumor.

Inclusion criteria: 1) Patients with skin tumors surgically resected and confirmed by postoperative pathological diagnosis; 2) Complete high-frequency two-dimensional ultrasound images and superb microvascular imaging (SMI) images; 3) Clear images without interference from foreign objects or artefacts.

Exclusion criteria: 1) Patients with inconclusive postoperative pathological results; 2) Poor-quality ultrasound images, such as blurry images, artifacts, or incomplete tu-

mor display; (3) Lack of multi-plane SMI images or maximum two-dimensional ultrasound plane of the tumor.

Ultrasound image acquisition and descriptive semantic feature collection

All patients underwent standardized ultrasound scanning of skin tumors using a Canon Aplio i900 color Doppler ultrasound system equipped with an ultra-high-frequency transducer i24LX8 (8-24 MHz). Patients were positioned to display the tumor best, and an ultrasound physician performed multi-plane scanning of the skin tumors in two-dimensional and SMI modes, mostly under default instrument settings. The two-dimensional sonograms showing the maximum tumor diameter were stored in Digital Imaging and Communications in Medicine (DICOM) format for training the machine learning model. The SMI mode was used to observe tumor perfusion, and the vascular index was obtained from highly vascularized layers, with the optimal cutoff point for diagnosing malignant tumors calculated based on the Youden index. Two researchers initially evaluated ultrasound images and then confirmed by an experienced sonographer with over 20 years of experience, assessing four descriptive semantic features: internal echogenicity (homogeneous, heterogeneous), boundary clarity (clear, unclear), shape regularity (regular, irregular), and posterior acoustic characteristics (attenuation, non-attenuation).

Image segmentation and extraction of ultrasound-based radiomic features

The exported DICOM images were converted to NRRD format using 3D Slicer (version 4.10.2) to avoid potential mismatch issues between masks and original images during subsequent analysis. Manual delineation of regions of interest (ROIs) was performed using 3D Slicer, and an experienced sonographer with over 20 years of expertise evaluated the accuracy of tumor boundary delineation and checked reproducibility, with any discrepancies resolved by group discussion. Python (version 3.10.1) was used to read two-dimensional ultrasound images containing ROIs, and ultrasound-based radiomics features were extracted using the PyRadiomics (<https://pyradiomics.readthedocs.io>) package. Sets of ultrasound-based radiomics features included first-order features, two-dimensional (2D) shape features, and texture features extracted from the original and wavelet-transformed images. Four hundred and sixty-four ultrasound-based radiomics features were extracted from each two-dimensional image, all normalized using Z-score standardization.

Feature selection and machine learning model construction

In this study, demographic characteristics, vascular index on SMI, and descriptive semantic features were

collectively referred to as clinical features. The 156 lesions from 143 patients were stratified and randomly divided into a training cohort (111 lesions) and a validation cohort (45 lesions) at a ratio of 7:3, ensuring complete isolation between training and validation cohorts. Clinical feature selection was performed using univariate and multivariate logistic regression (LR) analyses to identify risk factors ($p < 0.05$) and independent risk factors ($p < 0.05$) for malignant skin tumors. Odds ratios (OR) and 95% confidence intervals (95% CI) were calculated.

Ultrasound-based radiomics features were selected using t-tests and Mann-Whitney U tests, retaining features significantly associated with malignant skin tumors ($p < 0.05$). To prevent overfitting and reduce the impact of collinearity on predictive results, the selected feature subset was subjected to Least Absolute Shrinkage and Selection Operator (LASSO) regression, with 10-fold cross-validation used to determine the optimal lambda value and select the best features associated with malignant tumors. The feature subset selected using the optimal lambda value was used to build machine learning models, including support vector machines (SVM), logistic regression (LR), light gradient boosting (LGB), Bayesian networks (BNB), and multi-layer perception (MLP). LR was used to construct both the clinical model and the combined model (clinical model combined with the machine learning model). Model performance was

evaluated using ROC curve analysis, and AUC values, accuracy, sensitivity, and specificity were calculated. The DeLong test was used to compare the predictive abilities of different models. A nomogram was constructed based on the LR results for the combined model. Statistical analyses were performed using SPSS (version 26.0) and Python (version 3.10.1). Continuous and quantitative variables were presented as mean $\pm$ standard deviation (SD) or M (P25, P75), while categorical variables were expressed as frequency (percentage). Statistical significance was set at $p < 0.05$.

Results

Clinical characteristics

A total of 143 patients (156 lesions) were included in this study, comprising 61 males and 82 females. Pathological diagnoses showed 108 benign lesions (including 20 epidermoid cysts, 8 compound nevus, 22 intradermal nevi, 10 seborrheic keratoses, and others such as lipomas, hemangiomas, and pilomatricomas), and 48 malignant lesions (16 basal cell carcinomas, 10 squamous cell carcinomas, 3 malignant melanomas, and others) (Table I). There were significant differences ($p < 0.01$) between benign and malignant groups regarding age, maximum diameter, internal echogenicity, boundary clarity, shape regularity, and vascular index on SMI, but not in gender,

Table I. Distribution of Benign and Malignant Lesions in this Retrospective Cohort (n=156)

Clinicopathological Distribution				
Benign (n=108)	Intradermal Nevus	22	Blue Nevus	3
	Epidermoid Cyst ^φ	21	Dermoid Cyst ^ψ	3
	Seborrheic Keratosis	10	Dermatofibroma	2
	Compound Nevus	8	Epithelioid Granuloma	2
	Lipoma	5	Pilomatricoma	2
	Tenosynovial Giant Cell Tumor	4	Schwannoma	2
	Junctional Nevus	3	Others ^η	18
	Hemangioma	3		
Malignant (n=48)	Basal cell carcinoma	20	Well-differentiated liposarcoma	1
	Squamous cell carcinoma ^ξ	12	Aggressive fibromatosis ^θ	1
	Malignant melanoma	4	Low-grade myxofibrosarcoma	1
	Bowen's disease	2	Undifferentiated pleomorphic sarcoma	1
	Cutaneous metastatic carcinoma	2	Sebaceous carcinoma	1
	High-grade sarcoma	2	Squamous cell carcinoma	1
	Spindle cell rhabdomyosarcoma	1		

φ: A subset of epidermoid cysts demonstrates secondary granulomatous inflammation, likely related to cyst rupture or foreign body reaction. ψ: Portions of dermal cysts exhibit granulomatous infiltrates. η: The lesions were primarily composed of benign mesenchymal-derived tumors or hyperplasias (e.g., glomus tumor, spindle cell hyperplasia), with a minor component of cutaneous adnexal tumors demonstrating follicular/infundibular differentiation and reactive hyperplasias; rarely, space-occupying lesions attributable to metabolic disorders (e.g., gouty tophi, amyloid deposits) were observed. θ: The lesion was conclusively classified as a spindle cell tumor-like proliferative lesion (aggressive fibromatosis with malignant potential). ξ: Squamous cell carcinomas with heterogeneous differentiation (G1-G3), including 2 cases with concomitant seborrheic keratosis and 1 case associated with actinic keratosis

location, or posterior acoustic characteristics (Table II). No statistical difference was observed between the training and validation cohorts (Table III).

Feature selection

Univariate and multivariate analyses of clinical features from the training cohort are presented in Table IV. Age, all descriptive semantic features, and the vascular index on SMI were identified as risk factors in univariate analysis ($p < 0.05$). Multivariate analysis indicated that age (OR=1.112, 95%CI: 1.057-1.170, $p < 0.001$), poorly defined margins (OR=6.821, 95% CI: 1.782-26.110, $p = 0.005$), and irregular shape (OR=8.911, 95%CI: 1.329-59.729, $p = 0.024$) were independent risk factors in the training cohort. A total of 464 radiomics features were extracted from two-dimensional ultrasound images,

including 90 first-order features, nine 2D shape features, 110 gray level co-occurrence matrix (GLCM) features, 70 gray level dependence matrix (GLDM) features, 80 gray level run length matrix (GLRLM) features, 80 gray level size zone matrix (GLSZM) features, and 25 neighborhood gray-tone difference matrix features (fig 1). After applying t-tests and Mann-Whitney U tests, only features with $p < 0.05$ were retained. Further dimensionality reduction of the selected subset was performed using LASSO regression, resulting in 10 ultrasound-based radiomics features and their respective weights (fig 2 and Table V).

Model development and evaluation

Five machine-learning algorithms were employed to develop various models based on the selected feature subset, optimizing internal parameters for satisfac-

Table II. Clinical Characteristics of patients in the benign and malignant subgroups.

Characteristics		Benign (N=108%)	Malignant (N=48%)	p_1
Gender				0.706
Male		46 (42.6%)	22 (45.8%)	
Female		62 (57.4%)	26 (54.2%)	
Part				0.062
Other parts		58 (53.7%)	18 (37.5%)	
Head and neck		50 (46.3%)	30 (62.5%)	
Age		52.0 (35.0, 66.8%)		<0.001
DSFUI				
MAX-D		12.0 (6.4, 21.0%)		<0.001
IEL	N	80 (74.1%)	19 (39.6%)	<0.001
	Y	28 (25.9%)	29 (60.4%)	
Poorly Defined Margin	N	96 (88.9%)	13 (27.1%)	<0.001
	Y	12 (11.1%)	35 (72.9%)	
Irregular Shape	N	101 (93.5%)	28 (58.3%)	<0.001
	Y	7 (6.5%)	20 (41.7%)	
Rear Echo Attenuation	N	81 (75.0%)	32 (66.7%)	0.282
	Y	27 (25.0%)	16 (33.3%)	
SMI Vascular Index	< 23.3	69 (63.9%)	14 (29.2%)	<0.001
	> 23.3	39 (36.1%)	34 (70.8%)	

DSFUI = Descriptive semantic features of ultrasound images. MAX-D = Maximum diameter. HEL = Inhomogeneous Echogenicity of lesions. Y = yes, meets the corresponding descriptive characteristics. N = no, contrary to the corresponding descriptive features.

Table III. Clinical Characteristics of patients in the training and validation cohorts

Cohort	Statistical Characteristics	DSFUI							
	Age	MAX-D	IEL	Poorly Defined Margin		Irregular Shape		SMI Vascular Index	
			N	Y	N	Y	N	Y	
Training			72	39	78	33	94	35	91 20
Validation			27	18	31	14	17	10	39 6
X^2 / U	-0.687	-0.256	0.327	0.327	0.029	0.029	1.067	1.067	0.506 0.506
p_2	0.492	0.798	0.568	0.568	0.865	0.865	0.302	0.302	0.477 0.477

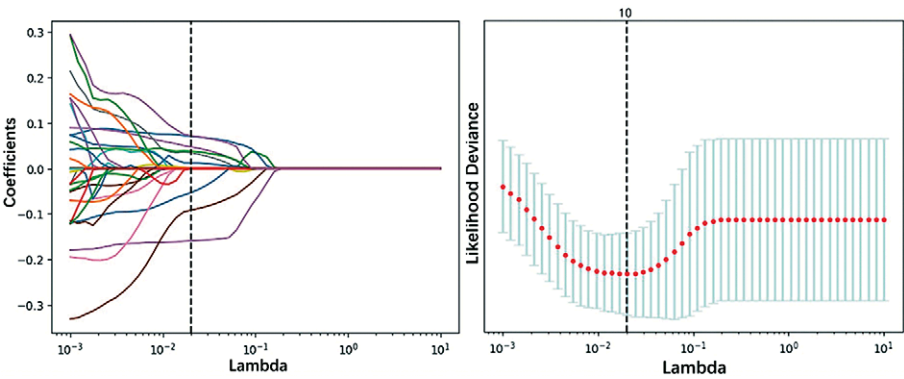


Fig 1. Cross-validated tuning parameters. Optimal LASSO tuning parameter values are indicated by dotted vertical lines; LASSO coefficient profiles of radiomics features. The coefficient profile plot was generated against the selected value using tenfold cross-validation; the vertical line was plotted with selected radiomics features.

tory performance (fig 3). Ten-fold cross-validation was used to determine the optimal lambda value and feature weights, allowing for the calculation of the radiomics score (Radscore). LR was used to construct clinical model and RWCM (Radiomics With Clinical Model) based on independent risk factors and the Radscore (fig 4), with a nomogram created for the combined predictive model (fig 5) [16]. The clinical model exhibited an AUC, accuracy, sensitivity, and specificity of 0.946, 0.865, 0.706, and 0.953 in the training cohort and 0.880, 0.844, 0.714, and 0.903 in the validation cohort. The MLP model outper-

formed other machine learning models, with higher AUC (0.964 and 0.894), accuracy (0.928 and 0.889), and specificity (0.907 and 0.968) compared to the clinical model. The RWCM’s AUC (0.953 and 0.885) was also superior to that of the clinical model (0.945 and 0.880) (Table VI). The results of DeLong’s test revealed a statistically significant discrepancy in efficacy between the MLP and clinical models ($Z=2.611$, $p=0.009$) (fig 6). Contrary to expectations, however, there was no statistically significant difference in performance between the RWCM and clinical models ($Z=0.068$, $p=0.946$) (Table VII).

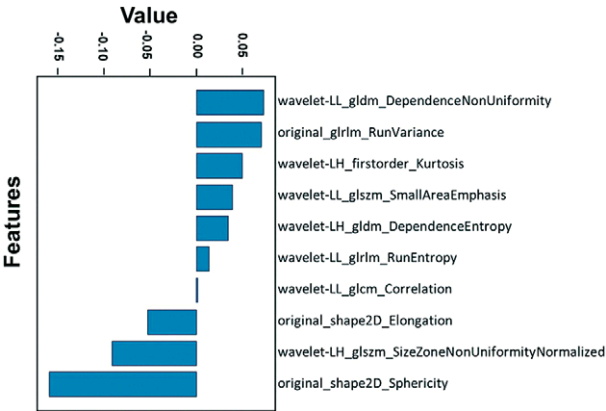


Fig 2. Coefficients of selected features.

Discussion

In this study, we constructed multiple models using ultrasound-based radiomics features and clinical features (statistical and descriptive semantic features of ultrasound images) to differentiate between benign and malignant skin tumors and evaluated their diagnostic performance. First, the ultrasound-based radiomics model trained with different machine-learning algorithms demonstrated effective classification of skin tumors, with the MLP model achieving an AUC of 0.912 in the validation cohort, outperforming other models and achieving comparable performance to the clinical model. It showed improved accuracy (0.889) and specificity (0.968) while

Table IV. Univariate and multivariate analyses of clinical features from the training cohort

	Univariate analyses		Multivariate analyses	
	OR (95% CI)	p	OR (95% CI)	p
Age	1.113 (1.075-1.153)	< 0.001	1.112 (1.057-1.170)	< 0.001
MAX-D	1.031 (1.007-1.055)	0.011		
IEL	4.361 (2.121-8.968)	< 0.001		
Poorly Defined Margin	21.538 (8.980-51.661)	< 0.001	6.821 (1.782-26.110)	0.005
Irregular Shape	10.306 (3.958-26.839)	< 0.001	8.911 (1.329-59.729)	0.024
SMI Vascular Index	2.792 (1.034-7.534)	0.043		

maintaining a high sensitivity. Second, age and two descriptive semantic features (poorly defined margins and irregular shape) were identified as independent risk factors for malignant skin tumors. Third, we established and

validated a novel predictive model using a nomogram to distinguish between benign and malignant skin tumors. The nomogram visually represents key factors affecting the outcome, simplifying the prediction of survival

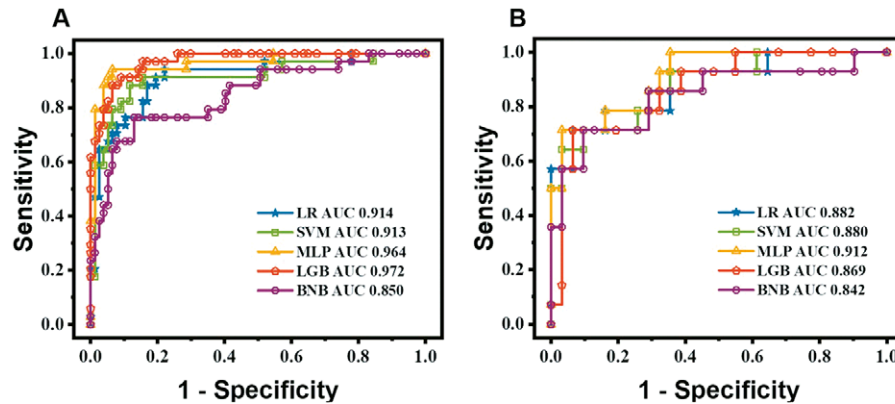


Fig 3. Receiver Operating Characteristic (ROC) curve for the performance of five machine learning models in the training group (A) and validation groups (B).

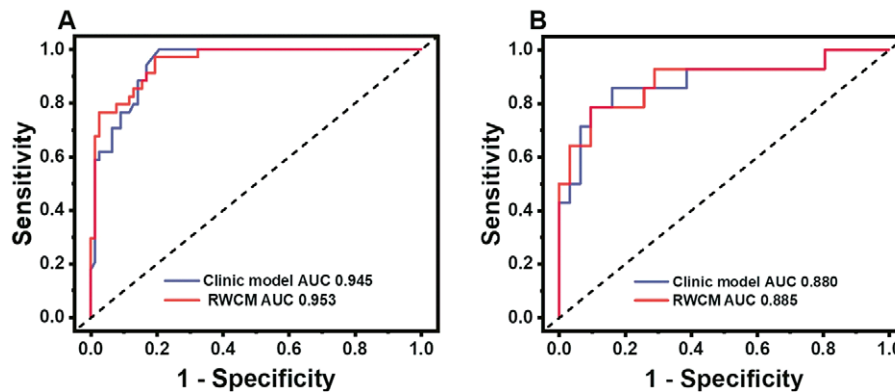


Fig 4. Receiver Operating Characteristic (ROC) curve for the performance of RWCM and clinical models in the training group (A) and validation groups (B).

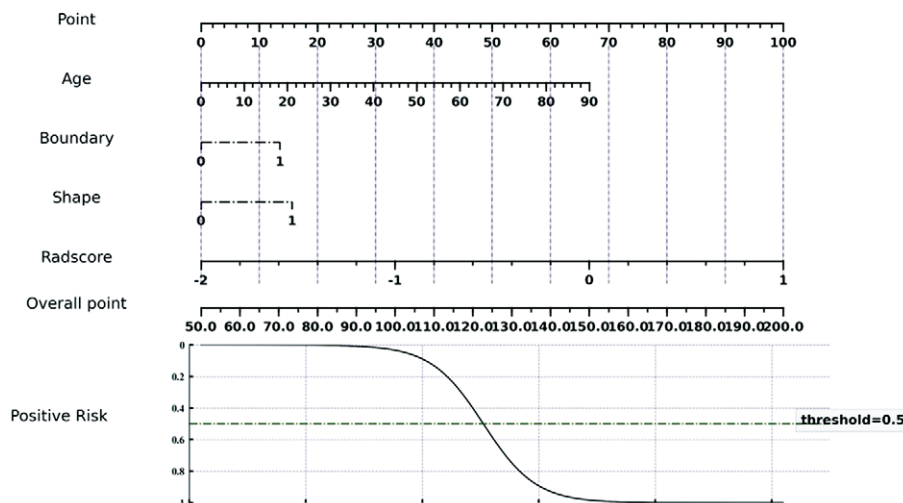


Fig 5. The RWCM nomogram. RWCM model combines age, boundary, shape and Radscore, developed in the training set.

Table V. Coefficients and Description of selected features.

Feature name	Description	Coefficients	
original_shape2D_Sphericity	Measure of the roundness of the shape of the tumor region relative to a circle.	0.15913	–
wavelet-LH_glszm_SizeZoneNonUniformityNormalized	Measures the variability of size zone volumes throughout the image.	0.09145	–
wavelet-LL_gldm_DependenceNonUniformity	Measures the similarity of dependence throughout the image.	0.071896	+
original_glrlm_RunVariance	Measure of the variance in runs for the run lengths.	0.070518	+
original_shape2D_Elongation	Elongation shows the relationship between the two largest principal components in the ROI shape.	0.0529	–
wavelet-LH_firstorder_Kurtosis	Measure of the ‘peakedness’ of the distribution of values in the image ROI.	0.047554	+
wavelet-LL_glszm_SmallAreaEmphasis	Measure of the distribution of small size zones.	0.038113	+
wavelet-LH_gldm_DependenceEntropy	Unknow.	0.034309	+
wavelet-LL_glrlm_RunEntropy	Measures the uncertainty/randomness in the distribution of run lengths and gray levels.	0.012685	+
wavelet-LL_glem_Correlation	Correlation shows the linear dependency of gray level values to their respective voxels.	0.000499	+
Lambda		0.306306306	

Positive sign (+) means proportional relationship, and negative sign (–) means inverse relationship.

Table VI. The AUC, accuracy, sensitivity, specificity and F1 score of five machine learning models, clinical model, and RWCM evaluated.

Cohort	Model	ACC	AUC	Sensitivity	Specificity	F1
Training	LR	0.865	0.914	0.706	0.953	0.762
	SVM	0.874	0.913	0.706	0.948	0.774
	LGB	0.884	0.972	0.912	0.857	0.816
	BNB	0.829	0.850	0.588	0.935	0.678
	MLP	0.928	0.963	0.853	0.961	0.879
	Clinical	0.865	0.945	0.706	0.953	0.762
	RWCM	0.892	0.952	0.765	0.948	0.812
Validation	LR	0.867	0.882	0.714	0.953	0.769
	SVM	0.867	0.880	0.643	0.968	0.750
	LGB	0.809	0.869	0.714	0.903	0.741
	BNB	0.844	0.842	0.571	0.968	0.696
	MLP	0.889	0.912	0.714	0.968	0.800
	Clinical	0.844	0.880	0.714	0.903	0.741
	RWCM	0.844	0.885	0.786	0.871	0.759

RWCM = Radiomics with Clinical Model.

probability through graphical representation. We combined clinical and ultrasound-based radiomics features using the nomogram, demonstrating a higher AUC in the validation cohort than in individual models [17,18]. Although the performance improvement was not statistically significant, the high sensitivity and moderate specificity of this model can help clinicians focus on suspicious lesions, which may require further examination to reduce the risk of missed diagnoses. Thus, the combined predictive model may have potential clinical utility.

Table VII. DeLong’s test in validation cohort.

Model	Z value	p
Clinical vs. MLP	2.611	0.009
Clinical vs. RWCM	0.068	0.946

Ultrasound-based radiomics refers to the quantitative analysis of multiple non-visually perceivable features in ultrasound images of lesions, transforming high-order features into data processable by computer programs, thereby enhancing the diagnostic and predictive role of

imaging [19]. Previous studies have demonstrated that models trained on clinical, dermoscopic, and histological images for skin tumor classification (benign/malignant) can achieve an AUC comparable to or better than that of expert readers [20,21]. The predictive ability of high-order features from ultrasound images has been well-established in benign and malignant breast and thyroid nodules [22]. However, studies that classify skin tumors based on ultrasound radiomics using multiple machine-learning models are limited. We utilized multiple machine-learning algorithms, including LR, Light GB, SVM, BNB, and MLP, to explore the classification capability of different models. The MLP model was adapted to the small dataset by adjusting the number of hidden layers and neurons. Among the ten selected ultrasound-based radiomics features, the highest weight was assigned to the original_shape2D_Sphericity, suggesting that shape characteristics from 2D ultrasound images play a key role in distinguishing skin tumors, such as the spindle-shaped hypoechoic appearance typical of melanoma [23,24]. However, combining the traditional clinical model with the ultrasound-based radiomics model did not lead to significant performance improvement, likely due to overlapping features between the subsets and the small sample size in this study, which complicates the classification of skin tumors. In typical clinical practice, ultrasound physicians do not rely solely on the interpretation of 2D images but integrate information from other imaging techniques, such as ultrasound Doppler, which is heavily dependent on the physician's skills and experience. Additionally, visually evident factors (tumor macroscopic appearance, tumor surface, and surrounding skin status) influence the physician's judgment. Haenssle et al demonstrated that a convolutional neural network (CNN) model trained on dermoscopic images outperformed dermatologists in distinguishing melanoma from benign nevi, even when clinical information was included [25]. In this study, we aimed to simulate the basic diagnostic capabilities of ultrasound physicians, which are inherently limited. However, the findings suggest that the combined analysis of 2D ultrasound and SMI imaging may provide more objective and valuable information in the future.

Elastography can facilitate quantitative analysis of parameters associated with benign and malignant skin lesions. Some researchers have integrated ultrasound elastography with the strain ratio, which compares the average strain value of normal skin to that of the lesion's region exhibiting the minimum strain. Their findings indicate that the elasticity of malignant skin tumors is reduced, aligning with previous research. This reduction in elasticity may be attributed to the increased cell density

within cancerous lesions, resulting in greater hardness and enhanced adhesion [26]. Additionally, we found that feature analysis of ultrasound strain elastography (USE) could be beneficial for quantitatively distinguishing cancerous from benign primary solitary skin lesions prior to biopsy [27]. This approach opens up new avenues for multimodal radiomics. By leveraging the image features from 2D imaging, superb microvascular imaging (SMI), and USE, we may be able to mitigate the effects of intra-tumor heterogeneity and feature overlap between different tumor types, thereby improving the accuracy of automatic diagnoses.

Several limitations must be noted in this study. First, the manually delineated region of interest (ROI) is inherently unstable, and image segmentation quality significantly impacts feature extraction. Second, retrospective studies are prone to selection bias and inherent error. Unfortunately, the subjects in this study were recruited from a single institution, resulting in a relatively small sample size. This limitation hinders the model's applicability to broader clinical settings and restricts the development of multi-classification tasks [28]. To better reflect real-world clinical scenarios, our study did not exclude any specific lesions. The limited sample encompasses not only various skin tumor types but also subtypes of certain tumors. This heterogeneity among multiple tumor types can lead to confusion in feature recognition, ultimately reducing classification accuracy [29]. As a result, when the clinical model and the combined model perform binary classification tasks, there may be incorrect predictions, and these errors are not limited to any specific lesion type. In future studies, we plan to increase the sample size and include multi-center data, while also utilizing neural networks that are better suited for complex challenges and multi-type image analysis. Additionally, we aim to design hierarchical classification models tailored to different tumor subtypes.

Furthermore, machine learning algorithms are often viewed as "black boxes," which means that the predictions generated by these models are often difficult to interpret [30]. However, certain texture features, such as Size-Zone Non-Uniformity Normalized (SZNN) and 2D Sphericity, hold significant importance. SZNN is a feature derived from the Gray-Level Size Zone Matrix (GLSZM) that measures the distribution differences among regions of varying sizes within the region of interest (ROI). A smaller Sphericity value indicates that the ROI deviates more from a circular shape.

In addition to these two features, we also identified Dependence Non-Uniformity, which is based on the Gray-Level Dependence Matrix (GLDM). This feature reflects the non-uniformity of dependence relationships

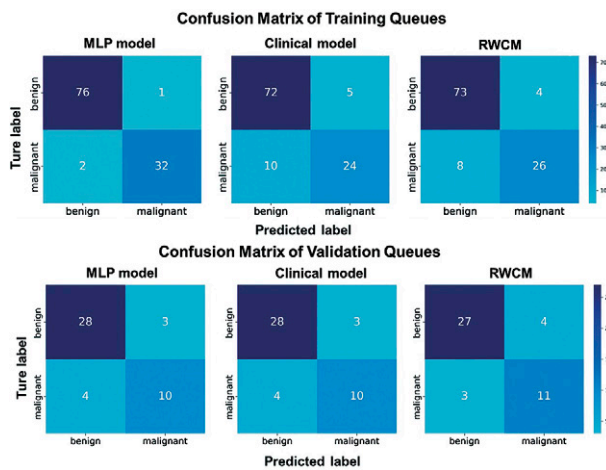


Fig 6. The confusion matrix of training and validation queues based on different model.

among gray levels in an image and characterizes the distribution of gray-level values within a lesion in various ways. Research has indicated that a higher number of hyperechoic spots is associated with the recurrence-prone micronodular, sclerosing variant, and morpheiform subtypes of basal cell carcinoma (BCC) [31]. Another relevant texture feature is High Gray-Level Emphasis, which falls under the GLDM category and measures the distribution of higher gray-level values. These features, along with their related metrics, may play a crucial role in helping machine learning algorithms differentiate BCC from other similar tumors [32,33], as well as distinguish between BCC subtypes. Effectively communicating how machine learning systems predict benign and malignant tumors to clinicians represents a vital area for future research.

Based on ultrasound deep learning models for improving the differential diagnosis of superficial soft tissue masses, this study employed a deep neural network to automatically extract medical image features, by passing the challenges of manually delineating tumor boundaries while focusing on changes in surrounding tissues and blood perfusion [34]. Preoperative ultrasound measurement of tumor thickness is an important parameter for surgical planning and prognosis prediction for skin cancers, particularly melanomas [35]. Differentiating tumor thickness is crucial in determining the safety margin for excision and sentinel lymph node biopsy during one-stage surgery. However, distinguishing lymphocytic infiltration or subcutaneous nevus cells from melanoma is challenging, often leading to an overestimation of tumor thickness [36,37]. Enhancing deep learning models to automate medical image feature extraction or progressively expand ROIs into deeper tumor regions could provide insights into the malignancy grading of skin tumors.

Conclusion

In conclusion, the performance of the machine learning model developed in this study matched the clinical model in distinguishing between benign and malignant skin tumors. These machine learning models may offer a potential non-invasive approach to improve diagnostic efficiency.

Conflict of interest: none

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